

PHYS 142/242 — Assignment 1

Harmonic Oscillator Path Integral

$$x_{\text{start}} = +0.890$$

All code is in `assignment1.ipynb` (Python/NumPy); running it top to bottom produces every figure, the animation `animation.gif`, the autograder file `results.json`, and the numbers quoted below. Units $m = \omega = \hbar = 1$, grid $x_i \in [-4, 4]$ with $N_D = 600$ ($\Delta x = 8/600$), $\epsilon = T_0/128$, $\alpha = 2$. One time step is $\psi_{n+1} = \Delta x \mathcal{K}_\epsilon \psi_n$, where $(\mathcal{K}_\epsilon)_{ba} = \mathcal{K}(x_b, \epsilon; x_a, 0)$ is the exact propagator evaluated on the grid.

Problem A: $\mathcal{K}_{8\epsilon}$

$\mathcal{K}_{8\epsilon} = (\Delta x)^7 \mathcal{K}_\epsilon^8$ (computed with `np.linalg.matrix_power`); `print(K_8eps)` gives

```
[[-0.05955846+0.0178667j -0.0757611 +0.0217024j -0.07967385+0.05414053j
... -0.46505982-0.30932634j -0.35439299-0.42735575j
-0.21687407-0.50917202j]
[-0.0757611 +0.0217024j -0.0896081 +0.03330227j -0.08308486+0.06724787j
... -0.53404108-0.16837676j -0.46141968-0.31131639j
-0.35439299-0.42735575j]
[-0.07967385+0.05414053j -0.08308486+0.06724787j -0.06226964+0.09303051j
... -0.56407376-0.01040964j -0.53404108-0.16837676j
-0.46505982-0.30932634j]
...
[-0.46505982-0.30932634j -0.53404108-0.16837676j -0.56407376-0.01040964j
... -0.06226964+0.09303051j -0.08308486+0.06724787j
-0.07967385+0.05414053j]
[-0.35439299-0.42735575j -0.46141968-0.31131639j -0.53404108-0.16837676j
... -0.08308486+0.06724787j -0.0896081 +0.03330227j
-0.0757611 +0.0217024j ]
[-0.21687407-0.50917202j -0.35439299-0.42735575j -0.46505982-0.30932634j
... -0.07967385+0.05414053j -0.0757611 +0.0217024j
-0.05955846+0.0178667j ]]
```

The matrix is complex symmetric, and also symmetric under $(x_a, x_b) \rightarrow (-x_a, -x_b)$, as the analytic formula requires. As a check, $\Delta x \mathcal{K}_{8\epsilon} \psi_0$ agrees with a single application of the exact propagator $\mathcal{K}(\cdot, T_0/16; \cdot, 0)$ to ψ_0 to within $1.3e-04$ (max. abs. difference). Individual matrix elements, however, differ from the exact kernel everywhere (by $\sim 10\text{--}20\%$ in the interior and $O(1)$ near the edges). The exact kernel is a rapidly oscillating distribution: the grid cannot represent its high-wavenumber part, and the intermediate integrals are cut off at ± 4 . Only the action of $\mathcal{K}_{8\epsilon}$ on smooth, localized states converges, and that is all the evolution needs.

Problem B: $\langle x \rangle(t)$

$\langle x \rangle = \Delta x \sum_i x_i |\psi_i|^2$. The state is not renormalized; its norm $\mathcal{N} = \Delta x \sum_i |\psi_i|^2$ stays in $[0.99940, 1]$. The numerical result lies on top of the classical trajectory $x_{\text{start}} \cos t$ (Ehrenfest's theorem).

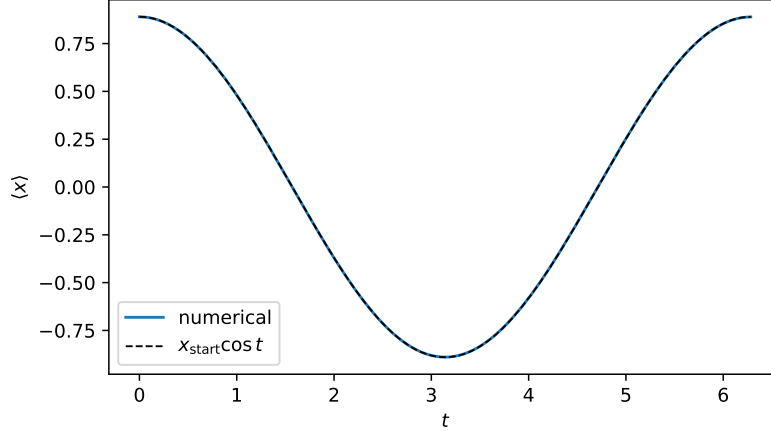


Figure 1: Mean position over one period; dashed line is $x_{\text{start}} \cos t$.

Problem C: $\langle E \rangle$, $\langle K \rangle$, $\langle V \rangle$

$\langle V \rangle = \Delta x \sum_i \frac{1}{2} x_i^2 |\psi_i|^2$ and $\langle K \rangle = \frac{1}{2} \Delta x \sum_i |\partial_x \psi|^2$, with $\partial_x \psi$ from central differences (`np.gradient`). $\langle E \rangle$ is conserved: it stays in $[1.0179, 1.0214]$, compared with the exact value $\langle E \rangle = \frac{1}{4}(\alpha + \alpha^{-1}) + \frac{1}{2} x_{\text{start}}^2 = 1.0211$. The small downward drift comes mainly from cutting the x integrals off at the box edge ± 4 , which makes the one-step matrix slightly non-unitary. It shows up as the same size error in $\langle V \rangle$, which uses no derivative, and it disappears on a larger box with the same Δx . The finite-difference derivative contributes only $\sim 10^{-4}$. The norm loss ($\mathcal{N} \geq 0.99940$, another symptom of the same truncation) lowers $\langle E \rangle$ by at most 0.06%, since the state is not renormalized. $\langle K \rangle$ and $\langle V \rangle$ exchange energy at frequency 2ω . Analytically

$$\langle V \rangle = \frac{1}{2} x_{\text{start}}^2 \cos^2 t + \frac{1}{4} \left(\alpha^{-1} \cos^2 t + \alpha \sin^2 t \right), \quad \langle K \rangle = \langle E \rangle - \langle V \rangle.$$

Two terms oscillate here. The first, $\frac{1}{2} x_{\text{start}}^2 \cos^2 t$, is the classical motion of the centre and is present even for a coherent state ($\alpha = 1$). The second comes from the width “breathing”, because $\alpha = 2$ makes the packet narrower than the ground state. They have opposite phase: $\langle V \rangle = \text{const} + [x_{\text{start}}^2/4 - (\alpha - \alpha^{-1})/8] \cos 2t$. For $x_{\text{start}} = +0.890$ they nearly cancel, which is why the $\langle K \rangle$ and $\langle V \rangle$ curves vary so little.

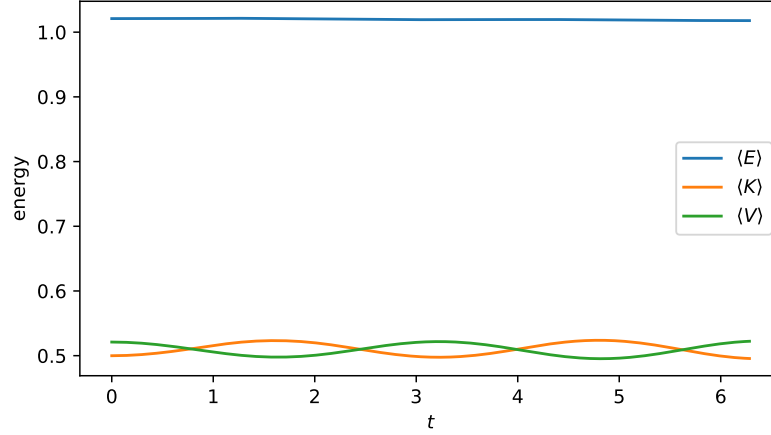


Figure 2: Mean total, kinetic and potential energy over one period.

Problem D: snapshots at $t = mT_0/16$

The packet moves from x_{start} to $-x_{\text{start}}$ in half a period. It is narrowest (tallest) at $t = 0$ and $T_0/2$ and widest at $t = T_0/4$, consistent with the width oscillation seen in Problem C.

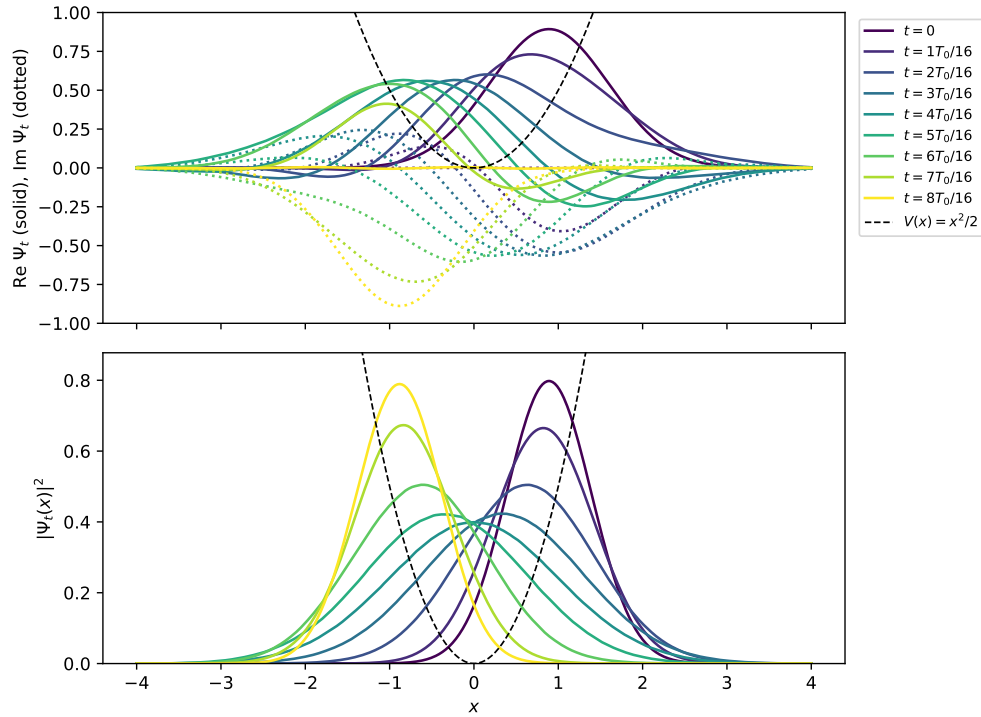


Figure 3: Top: amplitude $\text{Re } \Psi_t$ (solid) and $\text{Im } \Psi_t$ (dotted). Bottom: $P_t(x) = |\Psi_t(x)|^2$. Both at $t = mT_0/16$, $m = 0, \dots, 8$, with $V(x) = x^2/2$ superimposed.

Problem E: animation

`animation.gif` (submitted with the code) has 129 frames, one per step ϵ from $t = 0$ to T_0 . It shows $|\Psi|^2$, $\text{Re } \Psi$, $\text{Im } \Psi$ and $V(x)$. After one full period the state returns to Ψ_0 up to the global phase $e^{-i\pi}$, so $\text{Re } \Psi$ is flipped in the last frame.

Problem F (bonus): norm without renormalization

Prediction. The propagator is exact for any ϵ , so the only error is replacing $\int dx_a$ by $\Delta x \sum_a$. The phase of $\mathcal{K}_\epsilon(x_b, x_a)$ is $\phi = [(x_a^2 + x_b^2) \cos \epsilon - 2x_a x_b] / (2 \sin \epsilon)$, so as a function of x_a it oscillates with local wavenumber

$$k = \frac{\partial \phi}{\partial x_a} = \frac{x_a \cos \epsilon - x_b}{\sin \epsilon} \approx \frac{x_a - x_b}{\epsilon}.$$

The integral $\int dx_a \mathcal{K}_\epsilon(x_b, x_a) \psi(x_a)$ is dominated by the stationary point $k = 0$ ($x_a \approx x_b$), and elsewhere the fast oscillation cancels. On a grid, however, e^{ikx_a} and $e^{i(k-2\pi/\Delta x)x_a}$ are the same sequence. So the sum $\Delta x \sum_a$ also sees spurious “aliased” stationary points wherever $|k|\Delta x = 2\pi$, i.e. at $|x_a - x_b| \approx 2\pi\epsilon/\Delta x$. These give large, non-cancelling contributions. They stay outside the box, whose full width is 8, only if $2\pi\epsilon/\Delta x > 8$:

$$\epsilon > \epsilon_{\min} = \frac{4\Delta x}{\pi} = \frac{32}{\pi N_D}.$$

This depends only on the grid and the box, not on the initial state. For $N_D = 600$, $\epsilon_{\min} \approx 0.017 \approx T_0/370$; for $N_D = 150$, $\epsilon_{\min} \approx 0.068 \approx T_0/93$. So I predicted (i) $\epsilon = T_0/16 = 0.39$ works, while (ii) $\epsilon = T_0/512 = 0.012$ and (iii) $\epsilon = T_0/128$ with $N_D = 150$ both fail.

Result. After one period $\mathcal{N}(T_0) = 9.99 \times 10^{-1}$ for (i), 1.15×10^{258} for (ii), and 4.10×10^{51} for (iii). Cases (ii) and (iii) grow exponentially (straight lines on the log plot): when the kernel is undersampled, its aliased phase no longer cancels, the discrete one-step matrix $\Delta x \mathcal{K}_\epsilon$ is no longer (close to) unitary and has eigenvalues with $|\lambda| > 1$, so the norm is multiplied by a roughly constant factor each step. Counter-intuitively, the larger step (i) is the stable one: a small ϵ makes $\mathcal{K}_\epsilon$ oscillate faster, not slower.

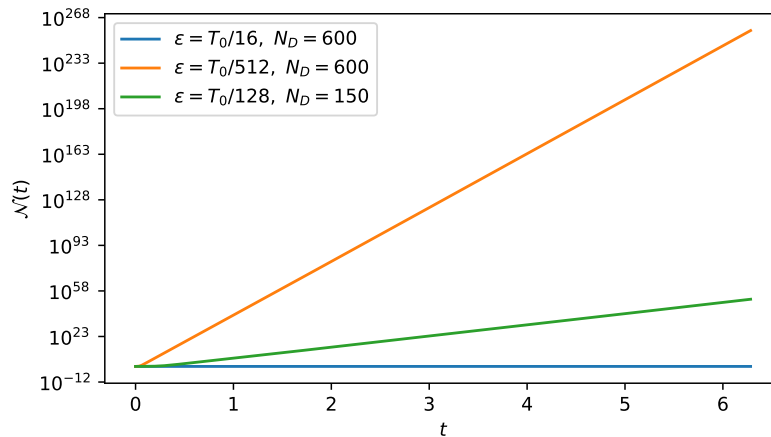


Figure 4: $\mathcal{N}(t)$ over one period for the three cases (log scale).

Smallest time step. Bisecting on $n = T_0/\epsilon$ (stable = $\mathcal{N}$ stays within 1% of 1 over one period) gives $\epsilon_{\min} = T_0/369 \approx 0.0170$ for $N_D = 600$, and $T_0/741 \approx 0.0085$ for $N_D = 1200$, in agreement with the estimate $32/(\pi N_D)$. The box-size dependence is confirmed too: at the same Δx on $[-8, 8]$ the threshold doubles to $T_0/187$, and it is unchanged for $x_{\text{start}} = 0$. Doubling N_D halves Δx and therefore halves the smallest usable time step: $\epsilon_{\min} \propto \Delta x \propto 1/N_D$.